



Nebraska Public Power District

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10 CFR 50.73

NLS2020061
October 15, 2020

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2020-002-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2020-002-00.

There are no regulatory commitments contained in this request.

Sincerely,

John Dent, Jr.
Vice President and
Chief Nuclear Officer

/dv

Attachment: Licensee Event Report 2020-002-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via ICES entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

IE22
NRR



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R 3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-mv/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk and omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Cooper Nuclear Station

2. Docket Number

05000298

3. Page

1 OF 4

4. Title

Secondary Containment Differential Pressure Perturbation Results in Potential Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	06	2020	2020	- 002 -	00	10	15	2020	Facility Name	Docket Number
										05000
										05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ Other (Specify here, in Abstract, or in NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact

Linda Dewhirst, Regulatory Affairs and Compliance Manager

Phone Number (Include Area Code)

(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
X	NG	CDMP	C340	Y	B	NG	TBG		Y

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On October 6, 2020, a spike in Secondary Containment differential pressure occurred exceeding the Technical Specifications limit of -0.25 inch of water gauge causing Secondary Containment to be inoperable for 3 minutes. The spike was due to a Reactor Building heating, ventilation, and air conditioning vortex damper that was binding and had an air leak in an associated instrument tubing.

Causal factors were an elevated operating force necessary to reposition the vortex damper and an air-line leak that developed due to contact between instrument tubing and a support brace.

The instrument tubing for the vortex damper was replaced and rubber material placed to prevent contact with the support brace. In addition, action will be taken to reroute the instrument tubing. Maintenance was performed to reduce the closing forces of the vortex damper. Maximum acceptable opening and closing forces will be determined and maintenance plans updated. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R 3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO
Cooper Nuclear Station	05000-298	2020	- 002	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at 100 percent power at the time of the event on August 6, 2020.

BACKGROUND

Secondary Containment serves as a barrier to confine and monitor potential releases during fuel handling operations and is a system that limits the release of radioactive materials to the environment and consists of four subsystems. One of the subsystems is the Reactor Building [EIS: NG], which encloses the Reactor Pressure Vessel [EIS: RPV] and Primary Containment [EIS: NH].

The secondary containment area (most of the Reactor Building) has supply and exhaust ventilating systems. The supply system furnishes filtered 100% outdoor air to all floors of the building through ductwork. The supply unit has an operating and standby (100% capacity) fan with vortex dampers [CDMP] to provide regulation of air capacity. The fans will deenergize in the event of loss of offsite power. The exhaust air is induced from the ventilated areas to a common plenum connected to the two exhaust fans, each of 100% capacity. The air is then exhausted to the atmosphere.

During normal plant operation, a minimum average negative pressure of 0.25 inches water gauge (wg) is maintained by differential pressure controllers which receive signals proportional to the pressure difference between outside air and the secondary containment atmosphere, and control the position of the exhaust fan vortex dampers. The differential pressure is detected by four separate probes, with one on each side of the Reactor Building. Control action is initiated from the average value of the four sensor probes.

If a Loss of Coolant Accident should occur, all ventilation systems of the primary containment area and secondary containment area will be isolated automatically, and the Standby Gas Treatment system will be automatically initiated.

EVENT DESCRIPTION

On August 6, 2020, at 1355, the Control Room received a computer alarm for Reactor Building differential pressure exceeding -0.25 inches wg and a change was noted on the Reactor Building differential pressure recorder. Personnel were dispatched to walk down the Reactor Building ventilation and Reactor Recirculation Motor Generator ventilation systems. No equipment deficiencies were identified during the walkdowns and Reactor Building differential pressure returned to normal without operator action after approximately three minutes. Operations personnel reviewed meteorological data including wind speed, wind direction and barometric pressures and concluded meteorological changes were enough to cause the momentary transient. Technical Specifications Surveillance Requirement (SR) 3.6.4.1.1 basis considers momentary transient due to gusty wind conditions acceptable and are not a cause for failure to meet the Surveillance Requirement.

On August 13, 2020, at 1500, Operations personnel identified an unexplained rise in Reactor Building pressure. Pressure rose from -0.34 inches wg to just below -0.25 inches wg. A review of this event

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CONTINUATION SHEET**

(See NUREG-1022, R.3 for Instruction and guidance for completing this form
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concluded it was not the result of meteorological conditions, but the Technical Specifications SR limit of -0.25 inches wg was not exceeded. The unexplained pressure transient was documented in the corrective action program, but Secondary Containment was not declared inoperable since -0.25 inches wg was not exceeded.

Engineering was asked to review the August 13th event and concurred it was the result of equipment performance not environmental conditions. Engineering was then asked to review the August 6th event and concluded that event was also the result of an equipment issue rather than meteorological conditions. A condition report was generated to document that Secondary Containment pressure exceeded -0.25 inches wg.

During the initial investigation of the condition, it was identified a change in operating force occurred in the spring of 2018 following repairs to the vortex damper. The required opening force measured during the annual preventive maintenance increased from 45 pounds to 69 pounds, while the required closing force increased from 21 pounds to 57 pounds. Following the rebuild, the controller output which had been relatively smooth with no sudden demand changes shifted to more erratic control. The change in demand characteristics is likely the result of the higher operating forces necessary to correctly position the vortex damper.

Subsequent troubleshooting identified binding of a vortex damper. Maintenance was performed to address the binding, with no noticeable binding when the damper was cycled following the maintenance. During the performance of post work testing, an air leak was identified in the copper tubing between the valve positioner and the lower diaphragm. The air leak developed due to contact between the tubing and a support brace and would prevent the normally applied air pressure from reaching the actuator. Maintenance was performed to replace the instrument line tubing.

BASIS FOR REPORT

This event is reportable as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident. The secondary containment was operable at the time of discovery; therefore, it was not reportable under 10 CFR 50.72(b)(3)(v) as an 8 Hr ENS Notification.

An engineering review was performed and the results demonstrate the perturbation on Secondary Containment pressure caused by the non-essential Reactor Building HVAC system did not impact the integrity of Secondary Containment and had no impact on the Standby Gas Treatment system. In the event of an accident, the Standby Gas Treatment system would have automatically started and would have been able to maintain vacuum on Secondary Containment as assumed in the Loss of Coolant Accident dose analysis. As such, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

SAFETY SIGNIFICANCE

This condition was non-consequential. The pressure excursion in the reactor building resulted in an unexpected plant condition and met the entry condition for required action pursuant to Technical Specifications. This event resulted in a minor excursion of Reactor Building differential pressure above the Technical Specifications limit of -0.25 inches wg for several minutes until the building HV controls restored the

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required differential pressure. This event did not negate the ability of the Standby Gas Treatment System to maintain Secondary Containment differential pressure, if required. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

CAUSE

The direct cause was the air-line leak between the positioner and the actuator that developed due to contact between the instrument tubing and a support brace. A second cause was the elevated operating force necessary to reposition the vortex damper.

CORRECTIVE ACTIONS

The instrument tubing for the vortex damper was replaced and rubber material placed to prevent contact with the support brace. In addition, action will be taken to reroute the instrument tubing. Maintenance was performed to reduce the closing forces of the vortex damper. Maximum acceptable opening and closing forces will be determined and maintenance plans updated.

PREVIOUS EVENTS

On May 1, 2020, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently opened simultaneously. This was reported under LER 2020-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On August 8, 2019, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2019-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On January 6, 2014, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2014-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On September 10, 2012, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently opened simultaneously. This was reported under LER 2012-003-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.